

Prediction of Histological Inflammation Severity Using Machine Learning

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Abstract—Evaluating inflammation severity from histology slides is difficult and can vary between pathologists. This study explores whether machine learning can be used to predict inflammation severity using quantitative features extracted from histological tissue images. These features describe cell densities, ratios and spatial relationships across different tissue regions. The dataset includes 106 patients, each with repeated measurements collected over time. Each sample is labelled with an inflammation severity score ranging from 0 to 5. Two prediction tasks are examined: multiclass prediction of the full severity score, and binary classification of inflammation status, defined as scores below 3 versus scores of 3 or higher. Logistic Regression and Random Forest models are evaluated using five-fold GroupKFold cross-validation to ensure that samples from the same patient are not split across training and testing sets. Models are trained using both the full set of features and a reduced subset of up to ten features selected based on feature importance. For binary inflammation classification, Random Forest achieved the best results, with an accuracy of approximately 0.908 and a ROC AUC of up to 0.969 when using the top ten features. Logistic Regression showed slightly lower but comparable performance. These results suggest that machine learning methods can provide consistent and objective support for binary inflammation assessment, even when only a small number of features is used.

Index Terms—Group cross-validation, histopathology, inflammation severity, machine learning, multiclass and binary classification

I. INTRODUCTION

Accurate assessment of inflammation in histological tissue samples is important for clinical decisions and for tracking disease progression. In many inflammatory diseases, severity scores are assigned by pathologists who visually inspect tissue slides and evaluate disease-related structural features. Although this approach is widely used in clinical practice, it depends heavily on human judgement. Differences in experience, interpretation, and tissue preparation can lead to variation in severity scores between observers and across different time points.

Previous studies have shown that specific features seen in histological tissue samples, such as immune cell infiltration and tissue structure changes, are linked to disease severity and long-term outcomes in inflammatory bowel disease [6].

To reduce this variability, machine learning methods have increasingly been used in histopathology. These methods rely on quantitative features extracted from tissue images and aim

to provide more consistent and reproducible assessments of disease severity [1],[4],[5].

II. MATERIALS AND METHODS

A. Dataset Description

The dataset contains histological assessments from 106 patients who participated in a clinical study. Each patient can have up to three assessments taken at different time points: baseline, first trimester, and one-year follow-up. Some visits were missed, so not all patients have data at every time point, which results in missing entries in the dataset. Each row corresponds to a single assessment and is linked to a unique patient identifier (PatID).

Each assessment is assigned an inflammation severity score from 0 to 5 based on expert pathological review. For every assessment, 141 quantitative features are also provided. These features capture cell densities, ratios between different cell types, and measures of how cells are spatially distributed within predefined tissue regions.

Assessments with missing severity scores were removed, while all remaining data were used for model training and evaluation.

B. Feature Representation

The input features are numerical values that describe the structure of the tissue at a microscopic level. They include measurements of cell density for different immune and structural cell types across several tissue regions, ratios between selected cell populations, and colocalisation features that describe how cells are spatially arranged at different distances from one another. Together, these features form a high-dimensional description of inflammation within the tissue.

The patient identifier (PatID) and the inflammation severity score were removed from the feature matrix before model training to avoid information leakage.

C. Learning Tasks

Two prediction tasks were examined. In the first task, inflammation severity was treated as a multiclass classification problem, with the goal of predicting an integer severity score from 0 to 5.

In the second task, the problem was simplified to binary classification by grouping scores below 3 as non-inflamed and scores of 3 or higher as inflamed. This binary setup reflects a clinically meaningful threshold and allows model performance to be evaluated using sensitivity and specificity.

D. Pre-processing

Missing feature values were filled in using median imputation and applied separately within each training fold. Feature standardization was applied only to models that require scaled inputs, such as Logistic Regression, while tree-based models were used without scaling. All pre-processing steps were included in the model pipelines so that they were learned only from the training data in each cross-validation fold.

E. Classification Models

Two supervised machine learning models were evaluated: Logistic Regression and Random Forest.

Logistic Regression served as a linear baseline model, with class balancing applied to address imbalance in the target labels.

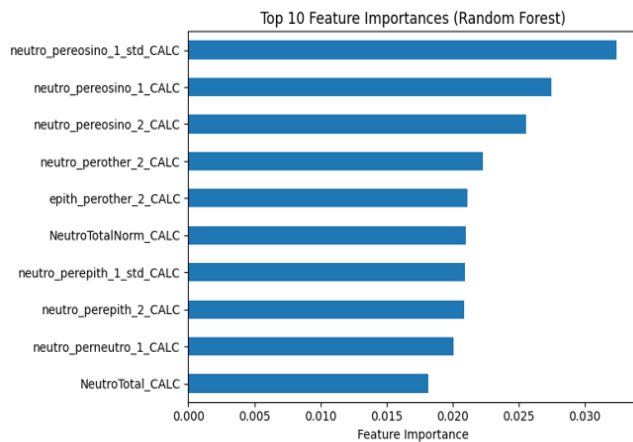
Random Forest was chosen as a non-linear ensemble method that can capture more complex relationships between features. Model hyperparameters were set in advance to avoid overfitting through excessive tuning.

F. Feature Selection

To study the impact of reducing the number of features, an additional experiment was conducted using no more than ten features.

Feature selection was carried out using a Random Forest model trained on the full feature set, and the top ten features were selected based on their feature importance scores.

These features were then used as input to both Logistic Regression and Random Forest models in subsequent experiments.



G. Validation Strategy and Performance Metrics

Model performance was evaluated using five-fold GroupKFold cross-validation. The patient identifier (PatID) was used to define groups, ensuring that all assessments from a single patient were assigned to the same fold. This prevents data leakage caused by repeated measurements from the same patient.

For the multiclass task, performance was reported using classification accuracy, summarized as the mean and standard deviation across folds. For the binary task, accuracy, sensitivity, specificity, and the area under the receiver operating characteristic curve (ROC AUC) were computed and averaged

across folds.

III. RESULTS

A. Multiclass Severity Prediction

Table I shows the performance of the models for multiclass prediction of inflammation severity scores (0–5). Results are summarized as the mean classification accuracy and standard deviation across five GroupKFold cross-validation folds. Performance is shown for both the full feature set and the reduced feature set containing the top ten features.

Table I

Multiclass classification accuracy (mean $\pm$ standard deviation)

Model	Feature Set	Accuracy Mean	Accuracy Std
Random Forest	All features	0.667	0.026
Random Forest	Top 10 features	0.627	0.018
Logistic Regression	Top 10 features	0.578	0.046
Logistic Regression	All features	0.553	0.067

B. Binary Inflammation Classification

Table II summarizes the performance of the models on the binary classification task, where severity scores were grouped into non-inflamed (score < 3) and inflamed (score $\geq$ 3) categories. Performance is reported using accuracy, sensitivity, specificity, and ROC AUC, averaged across cross-validation folds.

Table II

Binary classification performance

Model	Feature Set	Accuracy	Sensitivity	Specificity	ROC AUC
Random Forest	Top 10 features	0.908	0.930	0.872	0.969
Logistic Regression	Top 10 features	0.902	0.863	0.957	0.966
Random Forest	All features	0.912	0.909	0.912	0.958
Logistic Regression	All features	0.889	0.888	0.882	0.938

IV. DISCUSSION

The results show that Random Forest consistently performs better than Logistic Regression for both multiclass and binary classification tasks. This is expected, as ensemble methods such as Random Forest are well suited to high-dimensional biomedical data and can handle non-linear feature interactions and redundant variables effectively [2]. In contrast, Logistic Regression is a linear model and is therefore more limited in its ability to capture complex relationships between histological features.

This performance difference is linked to the nature of the feature space used in this study. The histological features describe complex cell distributions and spatial relationships with tissue regions, which are unlikely to be linearly separable.

As an ensemble of decision trees, Random Forest can capture these non-linear relationships more effectively than linear models, which helps explain their stronger performance across tasks.

Model performance differed between the two prediction tasks. For binary inflammation classification, models trained using only the top ten features achieved slightly better performance than models trained on the full feature set. In contrast, for multiclass severity prediction, the full feature achieved higher accuracy than the reduced feature set. This indicates that binary classification relies mainly on a small number of highly informative features, whereas predicting multiple severity grades benefits from a more detailed and comprehensive feature representation. Feature reduction can also reduce redundancy, improve interpretability, and lower computational cost, which is desirable for clinical use.

Overall performance was higher for the binary classification task than for multiclass severity prediction. From a clinical perspective, this is expected as separating tissue into non-inflamed and inflamed categories is a simpler decision than distinguishing between several adjacent severity grades, where overlap between classes is more likely. Receiver operating characteristic analysis is commonly used in clinical decision-making to assess the balance between sensitivity and specificity across different thresholds [3].

The high sensitivity and specificity achieved in the binary setting indicate that the proposed models can reliably identify clinically relevant inflammation while maintaining balanced error rates between classes.

Although confusion matrices are not explicitly shown, the reported sensitivity and specificity values suggest that models give different sensitivity and specificity trade-offs depending on the model and feature set. Such trade-offs may be important in clinical settings, where higher sensitivity may be preferred to minimize missed inflammation, or higher specificity may be favoured to reduce false positive diagnoses.

The use of GroupKFold cross-validation is important for the validity of these results. Since multiple assessments from the same patient are correlated, standard random cross-validation could allow patient-specific information to leak between training and testing sets, leading to overly optimistic performance estimates. By ensuring that all assessments from a single patient are kept within the same fold, the evaluation better reflects model performance on unseen patients and better represents real-world clinical deployment.

V. CONCLUSION

This study examined whether supervised machine learning can be used to predict inflammation severity from quantitative histological features. Both multiclass severity prediction and binary inflammation classification were evaluated using patient-level cross-validation to avoid data leakage from repeated measurements. Random Forest models consistently performed better than Logistic Regression, especially for the binary classification task. This reflects their ability to capture non-linear patterns in high-dimensional data.

Similar performance was obtained when using only a small set of informative features, showing that reliable inflammation

detection does not require the full feature set. This suggests that reduced feature representations may be sufficient for practical use while also improving interpretability and efficiency. Overall, these results indicate that machine learning methods, when evaluated using appropriate validation strategies, can provide robust and objective support for binary inflammation assessment. Future work could investigate ordinal models for severity grading and explore further improvements through feature selection or model refinement.

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